

FROM FRAGMENTED GROUND FLOORS TO A SHARED, CLIMATE-RESPONSIVE URBAN GRAMMAR



01 GARDEN  
ZHONGZHIYUAN / DAY  
ecology + research + low-disturbance testing



02 LEARNING  
AI ORIGIN / DUSK  
shared street + open ground floor + daily learning



03 STATION-CITY  
DAZHONGSI / NIGHT  
transfer + urban rooms + accessible public life

THE FIVE-BAND GRAMMAR

G0 MOVE

continuous public movement

G1 STAY

shade, ecology and comfort

G2 THRESHOLD

porch and transitional space

G3 SHARE

active and shared ground floor

G4 SERVICE

support, maintenance and AI

CORE POSITION

AI is an optional enhancement. Continuous, accessible and climate-responsive public ground remains the baseline.



GROUND-FLOOR OPPORTUNITY STRUCTURE

Ground-Floor Opportunity Map / 首层机会诊断图

DESIGN DIAGNOSIS PROVISIONAL GEOMETRY NOT TO SCALE

诊断首层界面活力与失衡，识别“加厚”空间机会，为五段厚基面语法与精细化干预提供依据。  
Diagnosing ground-floor conditions and imbalances to identify “thickening” opportunities and inform the five-band grammar and targeted interventions.

OPPORTUNITY TYPES / 机会类型

- 1 Existing Active Ground Floor 现状活跃界面  
现有连续商业、社区服务、开放入口等活跃界面。  
Active retail, services, open entrances.
- 2 Potential Opening / Retrofit Opportunity 开启 / 改造潜力  
封闭界面、可开启立面或改造潜力的底面空间。  
Blank walls, infill potential, retrofit openings.
- 3 Campus / Park Edge Interface 校园 / 公园边界界面  
校园、行政或大型公园与城市的过渡带。  
Edges of campuses, parks and institutional grounds.
- 4 Station-City Interface 站城界面  
轨道站点与城市公共领域的关键连接区。  
Key station areas where transit meets the city.
- 5 Blue-Green Edge Interface 蓝绿边界界面  
滨水、河道、绿道与城市的交汇地带。  
Rivers, canals, greenways meeting the city.
- 6 Adaptive Reuse Opportunity 适应性再利用机会  
低效工业、仓储、老旧建筑等可转化为公共功能。  
Underused industrial/warehouses, obsolete buildings.
- 7 Public-Service Interface 公共服务界面  
学校、医院、文化设施等与公共空间的对接区。  
Schools, hospitals, cultural & civic facilities.
- 8 Pedestrian Interruption / Weak Interface 行人中断 / 弱界面  
高架、围墙、空档面等造成的行人断点与弱界面。  
Barriers, walls, large setbacks, dead edges.

Existing Public / Open Ground 现状公共开放空间  
Blue-Green System 蓝绿系统 (河道 / 绿地 / 滨水)

CONCEPT DESIGN ON PROVISIONAL GEOMETRY  
NOT AN OFFICIAL REDLINE · OPPORTUNITY LOCATIONS REQUIRE PARCEL-LEVEL VERIFICATION



THE DIAGNOSTIC QUESTION

Where can the existing city ground become more continuous, open, shaded and shared?

- 1 BARRIERS  
weak crossings and interrupted routes
- 2 INACTIVE EDGES  
closed or inaccessible ground floors
- 3 CLIMATE GAPS  
missing shade, shelter and blue-green stay
- 4 SERVICE GAPS  
fragmented public-service connections

WHAT THIS PAGE CAN SUPPORT

- priority questions
- prototype selection

REQUIRED NEXT EVIDENCE

official GIS / ownership / access  
building condition / utilities / operations  
No scale, area or parcel claim is made here.

EVIDENCE DISCIPLINE

This page identifies design opportunities, not official parcels, precise areas or engineering boundaries.  
The opportunity structure must be rebuilt on official GIS before measurement or implementation decisions.



Thick-Ground Network / 厚基面网络总图

DESIGN NETWORK    PROVISIONAL GEOMETRY    CONCEPT PLAN

以基面条件为骨架，叠加五段空间干预，构建可步行、可停留、可共享、可服务的厚基面网络。  
A network of ground conditions and five-band interventions that builds a walkable, stayable, shareable, and serviceable thick-ground corridor.

GROUND-CONDITION FAMILIES / 基面条件谱系 (T1-T6)

- T1

Heritage Ground / 遗产型基面  
历史街巷、文物保护单位、文化场所周边。  
Historic lanes, heritage buildings and cultural precincts.
- T2

Campus Ground / 校园型基面  
高校校区、研究机构、学习场所周边。  
Universities, research institutes and learning interfaces.
- T3

Innovation Ground / 创新园区型基面  
科创园区、企业园、研发与试验场所。  
Innovation parks, enterprise campuses, R&D and maker facilities.
- T4

Neighborhood Ground / 社区型基面  
居住社区、生活服务、社区公共空间。  
Residential neighborhoods, daily services and community spaces.
- T5

Station-City Ground / 站城型基面  
轨道交通、综合体、交通换乘界面。  
Transit hubs, mixed-use complexes and multimodal interfaces.
- T6

Blue-Green Ground / 蓝绿型基面  
河道滨岸、绿地公园、生态与景观空间。  
Riverfronts, parks, ecological and landscape spaces.

FIVE-BAND INTERVENTIONS / 五段空间干预 (G0-G4)

- G0

MOVE

连续通行 / Continuous Movement

连续公共通行骨架与慢行优先网络。  
Continuous public movement spine and slow-priority network.
- G1

STAY

气候停留 / Climate Comfort

林荫与建筑、微气候遮蔽、停留与休憩点。  
Shaded and microclimate-comfort stay zones and pause points.
- G2

THRESHOLD

门脸界面 / Interface Edge

建筑前缘、入口门脸、软硬界面缓冲带。  
Porches, entry edges and soft-hard interface transitions.
- G3

SHARE

开放共享 / Open Clusters

广场、庭院、开放建筑、连接节点。  
Plazas, courtyards, active ground floors and ginkgocore spaces.
- G4

SERVICE

服务支撑 / Service Bands

后勤通道、设备带、物流与支撑空间。  
Service corridors, utility bands and support spaces.

ILLUSTRATIVE CROSS-CORRIDOR DEPLOYMENTS

←---→ S01-S12 are design-location IDs only.  
Their positions are provisional and require parcel, ownership, access, and field verification.

CONCEPT DESIGN ON PROVISIONAL GEOMETRY  
NOT AN OFFICIAL REDLINE · NETWORK ALIGNMENTS REQUIRE GIS/CAD REBUILD  
T1-T6 and G0-G4 are design taxonomies; parcel locations remain to verify.



CONCEPTUAL NETWORK DIAGRAM  
NOT TO SCALE

SIX DEPLOYMENT FAMILIES

- T1 Heritage ground
- T2 Campus ground
- T3 Innovation ground
- T4 Neighborhood ground
- T5 Station-city ground
- T6 Blue-green ground

THREE PROTOTYPE FAMILIES

GARDEN

ecology + research

LEARNING

street + open ground floor

STATION-CITY

transfer + urban rooms

NETWORK RULE

G0 remains continuous.  
AI remains reversible.

No verified dimensions or site boundaries.

Concept network only. Official GIS, boundaries and corridor alignments remain to verify.



THE FIVE BANDS

- G0

MOVE

continuous route
- G1

STAY

shade + ecology
- G2

EDGE

porous threshold
- G3

SHARE

active ground floor
- G4

SERVE

support + AI layer

BASELINE RULES

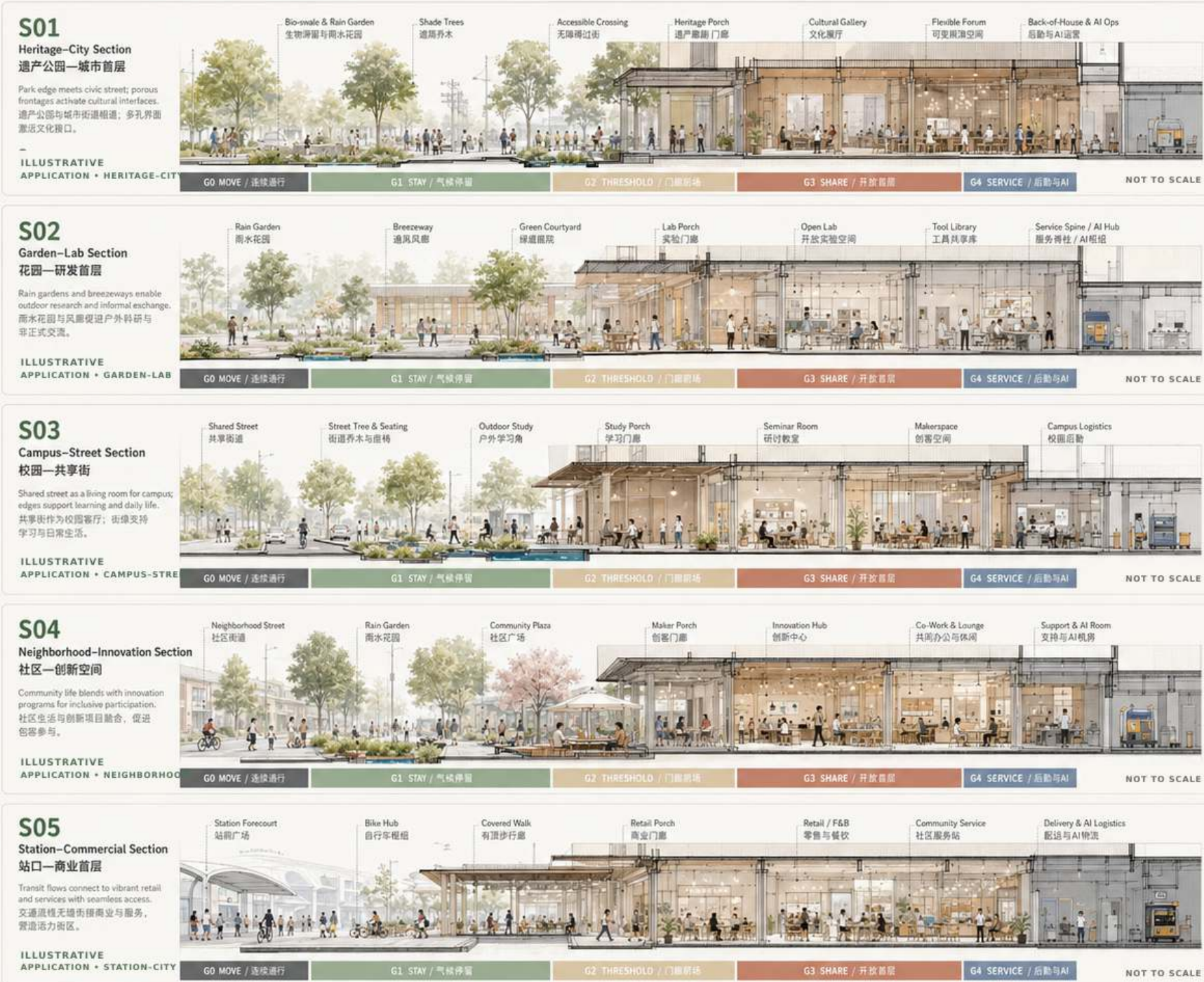
1. G0 stays open.
2. G1/G2 adapt.
3. G3 activates.
4. G4 enables.

DESIGN BOUNDARY

illustrative applications  
no verified dimensions  
sections require survey

Thick-Ground Section Atlas / 厚基面剖面图谱

Five-Band-Ground-Floor Grammar in Action / 五段空间语法的场景化表达





ZHONGZHIYUAN  
GARDEN THICK-GROUND  
CONCEPT PROTOTYPE

DESIGN PROVISIONAL NOT TO SCALE

花园里的研发，人与AI共同成长  
R&D in the Garden, Co-evolving with AI

在京张遗址公园与清河生态廊道之间，构建一个花园型创新区，将研发、测试、办公与公共生活融合在连续的绿色与开放的首层里，形成可漫步、可停留、可实验的AI创新生态。

图例 LEGEND

- 研发/办公建筑 R&D / Office
- 公共建筑 / 社区配套 Public Service
- 首层开放界面 Active Ground Floor
- 主要公共路径 Primary Public Path
- 次要步行路径 Secondary Path
- 自行车路径 Bicycle Path
- 入口 / 主要出入口 Main Entrance
- 地铁入口 Metro Entrance
- 地面停车 / 停车 Bicycle Parking
- AI测试区 AI Test Zone
- 雨水花园 / 生态滞留 Rain Garden
- 服务 / 后勤入口 Service Access

区位位置 LOCATION



设计策略 DESIGN STRATEGY

- 蓝绿渗透 Blue-Green Penetration
- 开放首层 Open Ground Floor
- 测试共生 Test in Public
- 慢行优先 Walk & Bike First
- 雨水韧性 Water Resilience
- 低碳智慧 Low-Carbon Smart

BASELINE EVIDENCE PANEL REMOVED

Replace with verified GIS / field survey:

- cadastral and ownership boundaries
- existing building footprints
- verified access and barrier conditions
- official blue-green infrastructure data

TO VERIFY

Excluded from A0 evidence claims until sourced.



重点剖面 KEY SECTIONS

A-A' 花园—研发剖面 Garden-Lab Section



B-B' 测试花园剖面 Test Garden Section



C-C' 滨水绿廊剖面 Waterfront Green Section



DESIGN PROTOTYPE · PROVISIONAL GEOMETRY · ALL DIMENSIONS AND SITE FACTS TO VERIFY

功能索引 PROGRAM INDEX

- 01 创新研发中心 R&D Center
- 02 AI联合实验室 AI Joint Lab
- 03 开放研发办公 Open Lab & Office
- 04 原型与验证中心 Prototyping Center
- 05 数据与算法中心 Data & Algorithm Hub
- 06 企业孵化器 Incubator
- 07 人才公寓 / 共享居住 Talent Housing
- 08 社区共享中心 Community Hub
- 09 运动与健康中心 Sports & Wellness
- 10 会议与展示中心 Conference & Expo
- 11 公共花园广场 Garden Plaza
- 12 AI测试花园 AI Test Garden
- 13 雨水花园群 Rain Garden System
- 14 服务与后勤中心 Service & Logistics
- 15 地铁出入口 Metro Entrance
- 16 自行车枢纽 Bike Hub
- 17 能源中心 Energy Center

流线分析 CIRCULATION



景观与生态结构 GREEN STRUCTURE



QUALITATIVE DESIGN INTENT

- public openness
- continuous walkability
- rainwater retention
- shaded outdoor comfort
- low-carbon operations

Quantified targets require verified baselines.

DESIGN PRIORITIES

1 GARDEN-LAB EDGE

research meets public landscape

2 BLUE-GREEN STAY

shade, rainwater and habitat

3 CLEAR GO ROUTE

continuous accessible movement

4 TEST GARDEN

public use remains the baseline

G0-G4 EMPHASIS

- G0 continuous garden walk
- G1 ecological stay space
- G2 lab-garden threshold
- G3 shared research ground
- G4 reversible test support

OPERATING RULE

Public access first.

Testing remains low-speed and reversible.

No verified scale, area or device commitment.

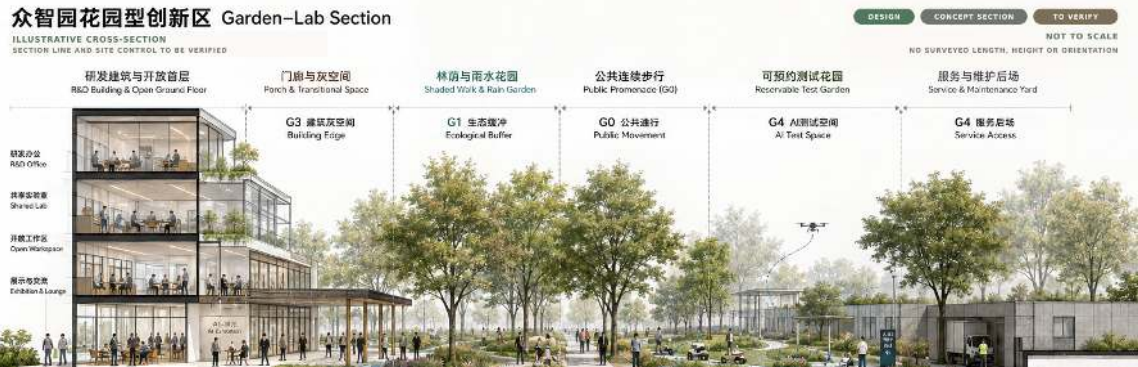
Concept prototype only. Official campus boundaries, access conditions and utilities remain to verify.



FLAGSHIP NODE



GARDEN-LAB SECTION



DAYTIME EXPERIENCE

garden walk + research encounter  
shade + play + low-speed testing

DESIGN PROTOCOL

1. G0 public movement remains uninterrupted.
2. Test zones are visible, bounded and reversible.
3. Non-digital wayfinding and staff support remain.
4. Climate comfort works without AI.

TO VERIFY BEFORE OPERATION

safety / liability / access hours  
maintenance / research ethics

Illustrative view and section. No operating commitment.



AI ORIGIN  
SHARED-STREET CONCEPT PROTOTYPE

DESIGN PROVISIONAL NOT TO SCALE

The learning-and-innovation thick-ground prototype—open ground floor, vibrant and shared, day and night.

- 1 开放首层网络 / Open Ground-Floor Network  
连续可渗透的G0空间，激活街道与庭院。  
Continuous permeable G0 layer activating streets and courtyards.
- 2 共享街道 / Shared Street  
慢行优先，共享通行，自行车友好，货运可达。  
Pedestrian priority, shared mode, bike-friendly, delivery-accessible.
- 3 学习庭院 / Learning Court  
户外学习与交流空间，连接教室、图书馆与廊下。  
Outdoor learning & exchange linking classrooms, library and arcades.
- 4 青年夜活力节点 / Youth Night Node  
夜间活动聚集地，灯光、演出、市集与社交。  
Night-activity hub with lighting, events, markets and social life.
- 5 门廊与过渡空间 / Threshold Porches  
深檐与门廊营造半开放界面，气候友好，延伸室内活动。  
Shaded porches and thresholds extend indoor life and support climate comfort.
- 6 共享庭院 / Shared Courtyard  
共创与休憩场所，绿植、座椅与可移动设施。  
Co-create and relax with greenery, seating and flexible furniture.
- 7 服务与后勤通道 / Service & Back-of-House  
二级后勤通道与后庭院，保障运营高效安静。  
Secondary service access and yards ensure efficient, quiet operations.
- 8 自行车与微交通 / Bike & Micromobility  
连续自行车网络，停放点与维修站点。  
Continuous bike network with parking and repair stations.
- 9 绿化与雨水管理 / Green & Rain Strategy  
雨水花园、渗透铺装与树荫廊道，改善微气候。  
Rain gardens, permeable paving and shaded corridors improve microclimate.
- 10 智慧与导航系统 / Wayfinding & Smart System  
统一导航、数据感知与交互设施。  
Unified wayfinding, sensing and digital interaction infrastructure.



ILLUSTRATIVE DESIGN LEGEND

- 共享街道 / Shared Street
- 慢行与步行主轴 / Pedestrian & Cycling Spine
- 自行车道 / Bike Lane
- 开放首层活动界面 / Active G0 Frontage
- 门廊与过渡空间 / Threshold Porches
- 共享庭院 / Shared Courtyard
- 口袋广场 / Pocket Plaza
- 学习庭院 / Learning Court
- 绿化树阵 / Planting & Tree Canopy
- 雨水花园 / Rain Garden
- 服务通道 / Service Access
- 二级后勤入口 / Secondary Service Entry
- 夜活力节点 / Night Node
- 导航与互动点 / Wayfinding Node
- 座椅与桌椅 / Seating & Furniture
- 自行车停放点 / Bike Parking
- 微交通站点 / Micro-Mobility Hub
- 公交站点 / Bus Stop

QUALITATIVE DESIGN INTENT

- open ground-floor network
- shared courtyards
- evening learning and social node
- shade and rainwater comfort
- bicycle and micro-mobility access

Targets require evidence before quantification

LOCATION DIAGRAM

Official base map and site boundary must be inserted before submission.

TO VERIFY

CONCEPT DIAGRAM  
NOT TO SCALE

METRIC SCALE REMOVED

SIX DESIGN MOVES

01 SHARED STREET  
walk, cycle, stay and meet

02 OPEN GROUND FLOOR  
learning, making and community use

03 LEARNING COURTYARD  
quiet work + informal exchange

04 BLUE-GREEN COMFORT  
shade, rainwater and seasonal use

05 MICRO-MOBILITY  
safe cycling and parking nodes

06 SERVICE SEPARATION  
deliveries and maintenance off G0

DAILY-LIFE BASELINE

The street works without an app.  
Ground-floor access is visible and legible.  
Human support remains available.

TO VERIFY  
ownership / fire access  
logistics / opening hours

No verified length, capacity or program target.

Concept prototype only. Building access, ownership, fire safety and service logistics remain to verify.



轴测、节点、剖面与暮景解释日常学习和创业生活 and community life share one porous street-ground interface

OPEN GROUND-FLOOR AXONOMETRIC

Open Ground-Floor Axonometric / 原点开放首层轴测

AI ORIGIN • CONCEPT PROTOTYPE  
ILLUSTRATIVE BLOCK • NOT A SURVEYED CENTRAL SECTION

以开放、共享、可逆的底层系统，激活学习与创业生态，白天服务学习与创新，夜间承载社区与青年生活。  
An open, shared, and reversible ground-floor system that activates learning and entrepreneurship ecosystems by day, and community and youth life by night.

1 开放前廊 / Open Porch  
灰空间过渡，内外可渗透。支持小型展陈与休憩。  
A shaded transition zone that blurs inside and outside; supports pop-up displays and seating.

2 共享街道 / Shared Street  
步行优先，低速共享，雨水花园与树阵提供舒适的微气候生态缓冲。  
Pedestrian priority street for walking, cycling and low-speed access, with rain gardens and trees for shade and ecological buffering.

3 活力首层 / Active Ground Floor  
学习、创业与服务功能沿街展开，室内外活动相互渗透。  
Learning, entrepreneurship and services activate street life and spill outdoors.

4 共享庭院 / Shared Courtyard  
半私密的学习与协作空间，连通室内外，可举办小型活动与市集。  
Semi-private courtyard for study, collaboration and events; connected indoors and out.

5 自行车停放 / Bike Parking  
便捷、安全的自行车停放与服务点，鼓励绿色出行。  
Convenient and secure bike parking and repair to encourage low-carbon travel.

6 后勤服务带 / Service Bay  
隐蔽的后勤与垃圾收集、设备管理通道，不干扰公共空间体验。  
Discreet back-of-house band for deliveries, waste collection and equipment access.

7 可逆AI层 / Reversible AI Layer  
设备与数据接口预留，支持活动感知、数字导览与互动体验，可按需启用。  
Reserved infrastructure and data layer for sensing, digital tools and interactive AI experiences; on-demand and reversible.

五层集成系统 / Five-Band Thick-Ground System



剖面示意 / Mini Section



DESIGN INTENT ON PROVISIONAL GEOMETRY  
PROGRAM, ACCESS, SERVICE AND AI INTERFACES REQUIRE SITE VERIFICATION.

CONCEPTUAL ORIENTATION  
NOT TO SCALE • DIMENSIONS AND NORTH TO VERIFY

SHARED-STREET SECTION

AI原点共享街精细剖面 Shared-Street Section

ILLUSTRATIVE CROSS-SECTION  
SECTION LINE TO BE VERIFIED



ILLUSTRATIVE KEY  
NO VERIFIED LOCATION  
NO SURVEYED ORIENTATION  
NOT TO SCALE

图例 LEGEND  
公共人行流线  
Public Pedestrian

骑行流线  
Bicycle Flow

服务物流线  
Service & Logistic

雨水流向  
Rainwater Flow

DIMENSIONS TO VERIFY

免透视  
k Perspective

ing, Innovation and Life

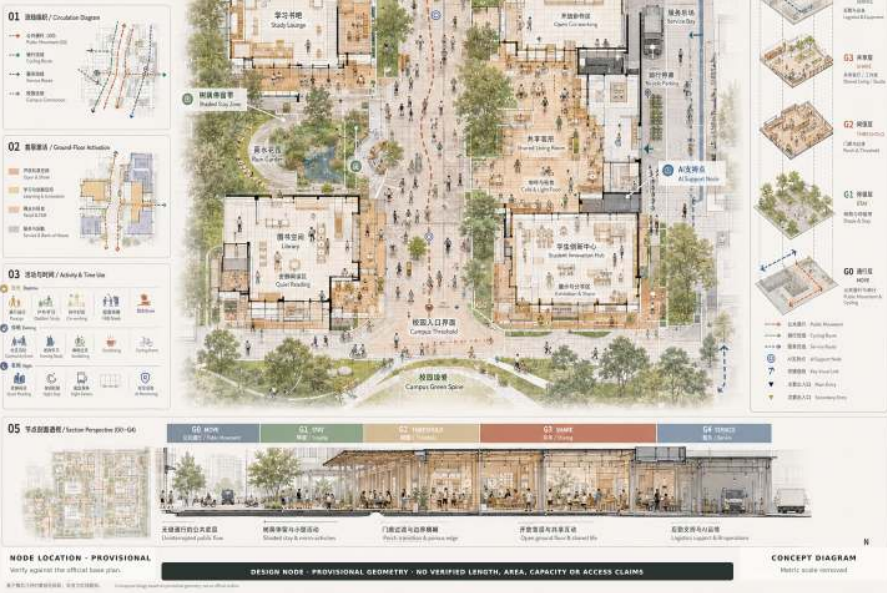


SHARED-STREET NODE

AI ORIGIN • SHARED STREET

CONCEPT NODE PROTOTYPE

DESIGN PROVISIONAL NOT TO SCALE

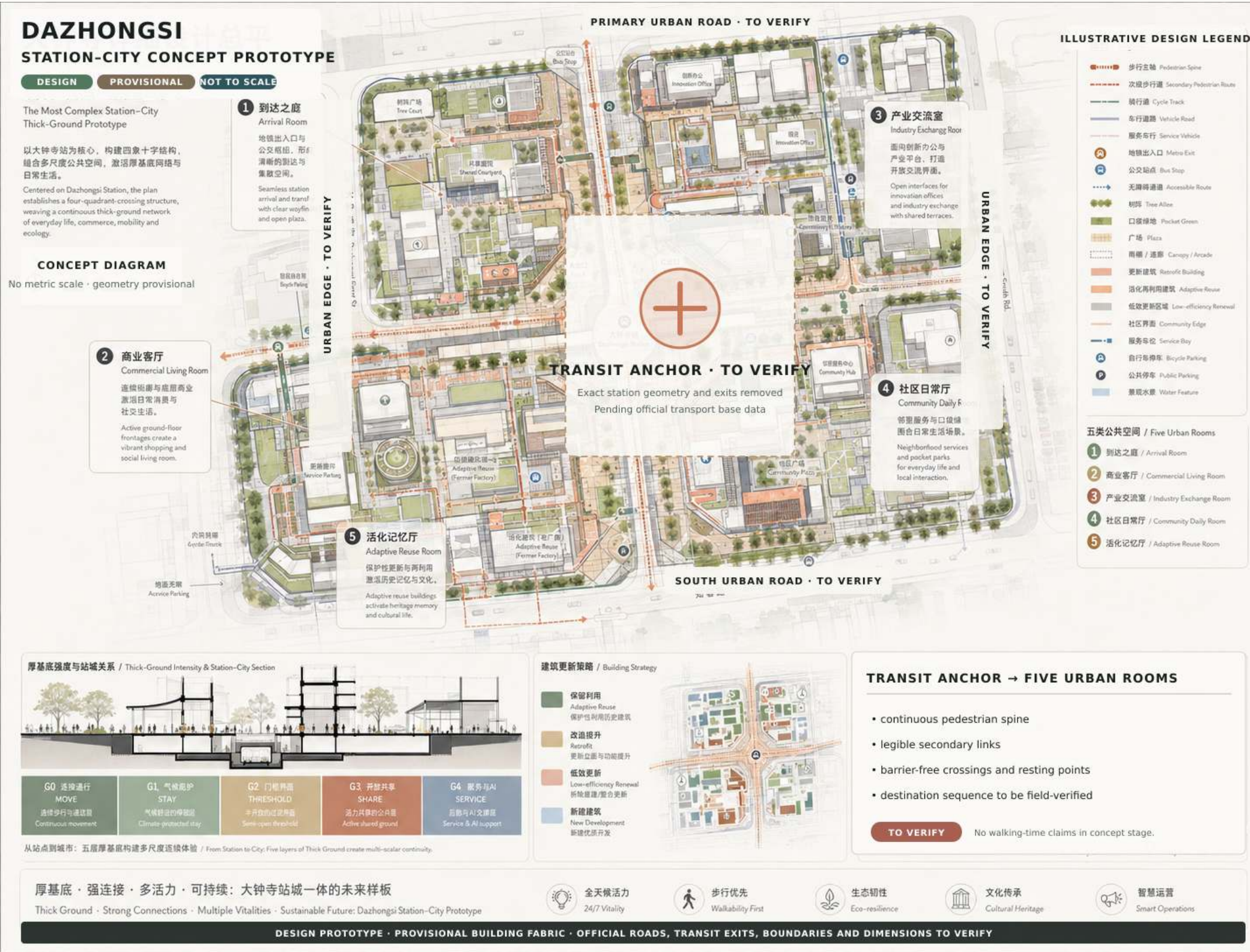


DUSK TRANSITION

research day -> learning + social  
+ community evening

Illustrative axon, node, section and view. No verified dimensions.





FIVE URBAN ROOMS

01 TRANSFER ROOM

clear arrival + accessible interchange

02 MEETING ROOM

waiting, meeting and public information

03 MARKET ROOM

active frontage + evening use

04 COMMUNITY ROOM

local access + everyday services

05 QUIET ROOM

rest, shade and lower-intensity stay

STATION-CITY RULES

G0 transfer remains direct.

G1/G2 create climate shelter.

G3 supports day-night public life.

G4 separates service and delivery.

TRANSPORT EVIDENCE

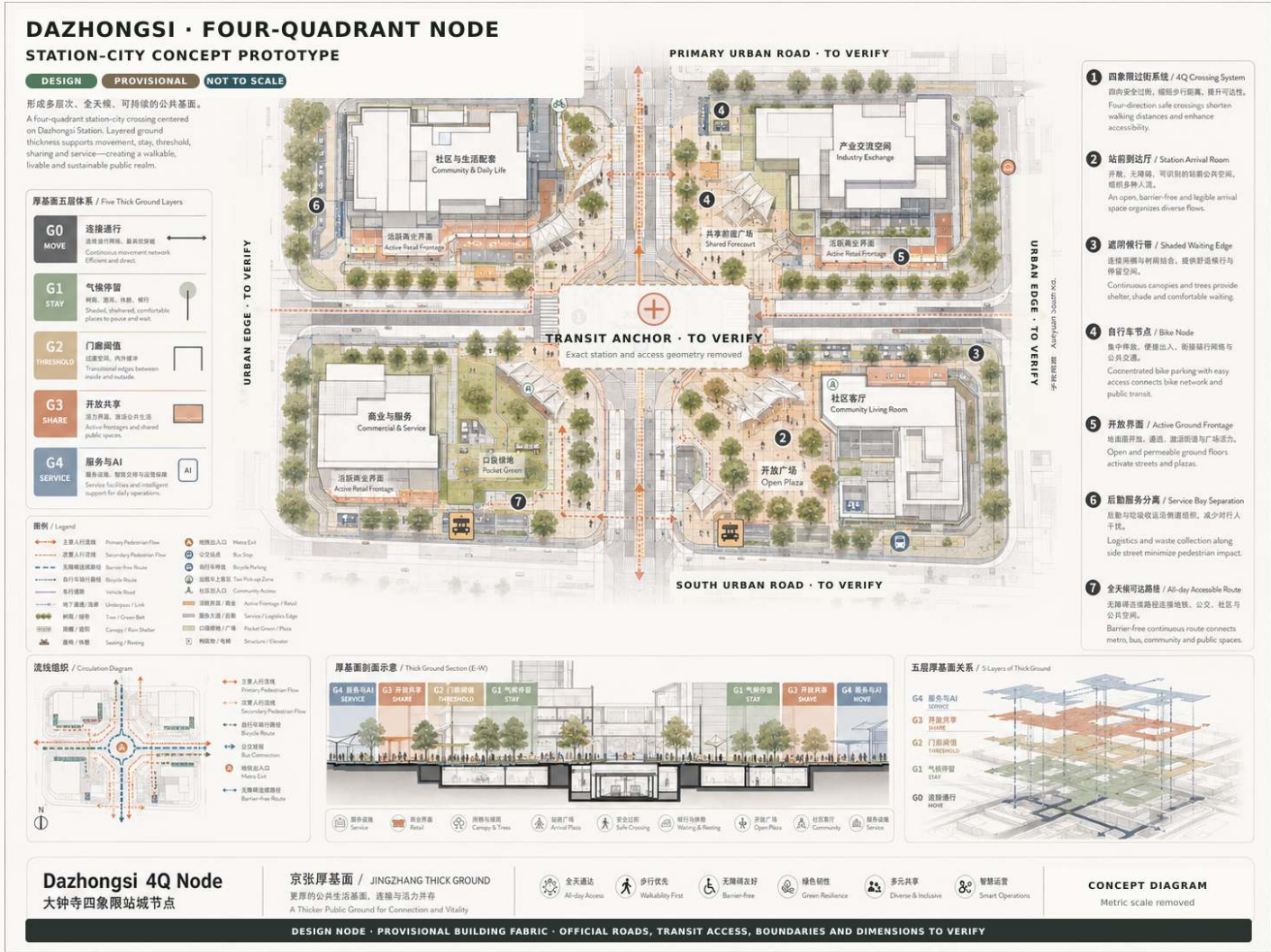
official entrances / lines / flows  
required before detailed design

No exit, line, flow or walking-time claim.

Concept prototype only. Official station entrances, lines, passenger flows and road conditions remain to verify.



FOUR-QUADRANT NODE



FOUR QUADRANTS

Q1 ARRIVAL

transfer + information

Q2 MEET

wait + gather

Q3 COMMUNITY

services + local access

Q4 MARKET

frontage + evening use

Quadrant locations are conceptual and require official station data.

STATION-CITY SECTION



Conceptual transfer stack. Underground lines, structures, entrances and technical clearances remain to verify.



24 HOURS x FOUR SEASONS

OPERATIONAL SUMMARY / A0 READING SCALE

DESIGN TOOL

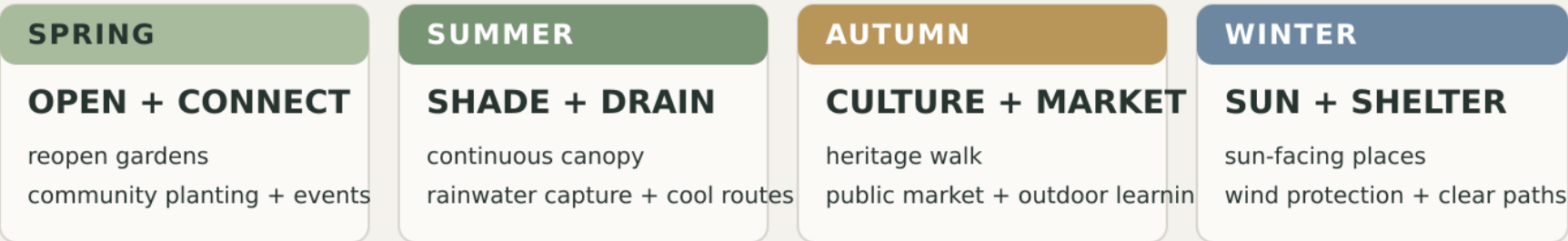
ILLUSTRATIVE

Times indicate a design scenario, not an operating commitment.

DAILY RHYTHM



SEASONAL PRINCIPLES



THICK-GROUND OPERATING LOGIC

GO ALWAYS OPEN -> G1/G2 ADAPT -> G3 ACTIVATES -> G4 ENABLES

CONCEPTUAL SCHEDULE / NO OPERATING COMMITMENT / VERIFY WITH USERS, MANAGEMENT AND CLIMATE DATA

Illustrative operating rhythm. Times indicate scenarios to test, not committed opening hours.

OPERATING PRINCIPLES

- G0 Always legible**  
continuous access + fallback
- G1 Climate responsive**  
shade, rain, sun, shelter
- G2 Threshold adaptable**  
open, pause, gather, close
- G3 Program reversible**  
learn, work, meet, rest
- G4 Service separated**  
maintain, deliver, secure

NON-AI BASELINE

Static signs, staffed help, manual management and safe public access

AI+ ENHANCEMENT

Optional sensing and coordination; never a condition for basic use.

VALIDATION LOOP

01 / OBSERVE

movement, dwell, comfort

02 / CO-DESIGN

users + operators + owners

03 / PILOT

reversible schedule + protocol

04 / ADAPT

seasonal review before scaling



THICK-GROUND COMPONENT KIT

FIVE REPRESENTATIVE COMPONENTS + THREE INTERVENTION LEVELS

A0 SUMMARY

TO VERIFY

REPRESENTATIVE KIT

TG-01 / G0

ACCESSIBLE WALK

continuous, clear, legible

TG-03 / G1

RAIN GARDEN

shade, habitat, water

TG-04 / G2

OPEN PORCH

threshold, shelter, pause

TG-05 / G3

ACTIVE GROUND

share, learn, meet, work

TG-11 / G4

SERVICE BAY

deliver, maintain, reverse

Full 13-component reference table moves to A3-P12.

INTERVENTION INTENSITY

L1 / LIGHT TOUCH

seating + shade  
wayfinding + paving repair  
quick, reversible, low-disruption

L2 / INTERFACE RETROFIT

ground floor + porch  
courtyard + crossing  
building-public realm coordination

L3 / STRUCTURAL CHANGE

station-city + adaptive reuse  
selective new + utilities  
conditional, evidence-led, phased

LEVEL DEPENDS ON CONDITION, OWNERSHIP, COST, APPROVAL AND OPERATING CAPACITY

CONCEPT TOOLKIT / SITE FIT AND TECHNICAL PERFORMANCE TO BE VERIFIED

Representative kit. Final dimensions, materials and performance depend on site and technical review.

FULL 13-COMPONENT INDEX

- TG-01 / G0

Accessible continuous walk
- TG-02 / G1

Shade + seating band
- TG-03 / G1

Rain garden
- TG-04 / G2

Open porch / threshold
- TG-05 / G3

Active ground floor
- TG-06 / G3

Shared courtyard
- TG-07 / G1+G3

Pocket urban room
- TG-08 / G0+G4

Cycle + micro-mobility
- TG-09 / G0+G2

Safe crossing
- TG-10 / G1+G3

AI Test Garden
- TG-11 / G4

Service / delivery bay
- TG-12 / G0+G2

Static + smart wayfinding
- TG-13 / G3+G4

Reversible AI interface

SELECTION RULE

Choose the smallest intervention that meets public value and safety needs.

AI interfaces remain optional and reversible.

INTERVENTION DECISION GATES

L1 / LIGHT TOUCH

quick, reversible, low disruption

L2 / INTERFACE RETROFIT

building-public realm coordination

L3 / STRUCTURAL CHANGE

conditional, evidence-led and phased



RETAIN -> RETROFIT -> REPAIR -> REUSE -> SELECTIVE NEW  
EVIDENCE-LED RENEWAL PATH / STRATEGY BEFORE PARCEL DECISIONS

A0 SUMMARY TO VERIFY

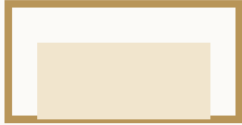
FIVE ACTIONS

01 / RETAIN



**KEEP WHAT WORKS**  
maintain sound buildings and useful public space

02 / RETROFIT



**OPEN THE INTERFACE**  
upgrade entrance, porch and active frontage

03 / REPAIR



**FIX PUBLIC SPACE**  
walk, shade, crossing and rainwater continuity

04 / REUSE



**ADAPT, DON'T ERASE**  
transform underused buildings for new use

05 / SELECTIVE NEW



**FILL VERIFIED GAPS**  
new only where structure and evidence require it

PHASED IMPLEMENTATION / NO FIXED DATES

1 QUICK REPAIR

survey + co-design  
quick, reversible actions

2 INTERFACE UPGRADE

ground-floor retrofit  
access + public-space repair

3 CONDITIONAL CHANGE

adaptive reuse  
selective new insertion

STRATEGY FIRST / INVENTORY MAP REMOVED FROM A0

Parcel decisions require ownership, condition, planning, cost and operating evidence.

CONCEPTUAL RENEWAL PATH / NO DEMOLITION OR DEVELOPMENT COMMITMENT

Strategy diagram only. No demolition, development, parcel or schedule commitment is implied.

PARCEL DECISION GATES

01 / CONDITION

structure, use and maintenance

02 / OWNERSHIP + RIGHTS

consent, access and responsibilities

03 / PUBLIC VALUE

access, inclusion and daily benefit

04 / TECHNICAL FIT

safety, utilities, climate and cost

05 / OPERATING CAPACITY

stewardship, funding and feedback

PRIORITY

Repair public ground and interfaces before considering structural change.

Formal inventory mapping follows verified data.

PHASED DELIVERY / NO FIXED DATES

1 QUICK REPAIR

survey + co-design + reversible action

2 INTERFACE UPGRADE

ground floor + access + public realm

3 CONDITIONAL CHANGE

reuse + selective new, only after gates



12 个 AI+ 城市生活场景卡 12 AI+ Urban-Life Scenario Cards

让算法服务生活，让城市更可达、更安全、更有温度  
AI Empowers Life, Making the City More Accessible, Safer and Warmer

厚基面层级 Thick Ground Levels (G0-G4)

G0 公共通行 Movement

G1 生态停留 Stay

G2 门庭过渡 Transition

G3 开放共享 Active

G4 服务支撑 Support

时段 Time of Day

日间 Day

傍晚 Evening

夜间 Night

全天 All Day

三大旗舰场景 Three Flagship Scenarios

SCN-01 众智园 AI Test Garden 众智园 Zhongzhiyuan

WHO 谁 Researchers, engineers, students, visitors, residents

WHERE 哪里 Central Garden & Test Courts, Zhongzhiyuan

WHEN 何时 日间为主，全天可达  
Primarily daytime, accessible all day

SPACE 空间 G1 / G2 / G3 (花园、门庭、共享空间)  
G1 / G2 / G3 (Garden, Transition, Active Space)

ACTION 做什么 休憩、讨论、观察测试、参与共创  
Rest, discuss, observe tests, co-create

AI+ 增强什么 AI+ Enhances 环境感知与舒适优化，机器人协同，导航环境，客流与安全监测  
Micro-climate optimization, robot coordination, smart wayfinding, crowd & safety monitoring

FALLBACK 没有AI怎么办 仍可使用花园，人工管理并基础设施保障  
Garden remains usable with manual management

TEST 如何验证 舒适度提升，参与人数，测试项目数量  
Comfort improvement, participation, # of tests

SCN-04 AI原点 Origin Shared Street AI原点 AI Origin

WHO 谁 大学生、创客团队、居民、上班族、亲子家庭  
Students, start-ups, residents, workers, families

WHERE 哪里 AI原点共享街与口袋广场  
Origin Shared Street & Pocket Plaza

WHEN 何时 午后—傍晚—夜间  
Afternoon—Evening—Night

SPACE 空间 G2 / G3 (过渡空间、开放空间、共享街道)  
G2 / G3 (Transition Space, Active Ground Floor, Street)

ACTION 做什么 学习、共创、咖啡社交、市集流动、夜间散步  
Study, co-create, cafe, markets, night stroll

AI+ 增强什么 AI+ Enhances 人流热力与活动推荐，空间感知，能耗优化，多感官交互与信息推送  
Crowd heat map & program recommendation, space booking, energy optimization, multi-sensory info

FALLBACK 没有AI怎么办 街道与广场仍可开放，信息通过传统方式获取  
Space stays open; info via signs and staff

TEST 如何验证 驻留时长，活动参与度，商户经营，能耗下降  
Dwell time, event participation, business vitality, energy saving

SCN-08 大钟寺 4Q Accessible Transfer 大钟寺 Dazhongsi 4Q Node

WHO 谁 通勤者、换乘旅客、居民、游客、老年人、残障人士  
Commuters, transfer passengers, residents, visitors, seniors, people with disabilities

WHERE 哪里 大钟寺地铁站及4Q城市客厅  
Dazhongsi Station & 4Q Urban Room

WHEN 何时 早晚高峰 + 全天候夜间  
Peak hours & all day to night

SPACE 空间 G0 / G2 / G3 / G4 (通行、过渡、开放、服务)  
G0 / G2 / G3 / G4 (Movement, Transition, Active, Support)

ACTION 做什么 高效换乘，接送全面，休憩、购物、办证服务，夜间社交  
Transfer, meet & greet, rest, shop, access services, night social

AI+ 增强什么 AI+ Enhances 客流预测与分流，无障碍路径导航，实时公交/地铁信息，应急与安全联动  
Passenger flow prediction, accessible routing, real-time transit info, emergency & safety

FALLBACK 没有AI怎么办 清晰标识与人工服务保障基本运行  
Clear signs and staff ensure basic operation

TEST 如何验证 换乘效率，无障碍通行时间，拥挤度下降，满意度提升  
Transfer efficiency, accessibility time, crowding reduction, satisfaction

SCN-02 研发午间共享院 Workday Lunch Courtyard G1 G3

WHO 谁 园区员工、周边居民  
Workers, nearby residents

WHERE 哪里 众智园共享院  
Zhongzhiyuan Shared Courtyard

WHEN 何时 午间  
Lunch time

SPACE 空间 AI+ AI+ 座椅与遮阳推荐，排队与配送优化  
Seat & shade suggestion, queue & delivery optimization

SCN-03 蓝绿舒适步行 Blue-Green Walk G0 G1

WHO 谁 行人、慢跑者、骑行者  
Pedestrians, joggers, cyclists

WHERE 哪里 三大节点之间的步行绿廊  
Green Corridors between Nodes

WHEN 何时 全天  
All day

SPACE 空间 AI+ AI+ 微气候指数提示，路线推荐，拥挤预警  
Micro-climate index, route rec., crowd warning

SCN-05 夜间学习街角 Night Learning Corner G2 G3

WHO 谁 学生、青年、自由职业者  
Students, young people, freelancers

WHERE 哪里 AI原点街角与口袋空间  
Origin Street Corners

WHEN 何时 夜间  
Night

SPACE 空间 AI+ AI+ 学习空间预约，噪音监测，照明自适应  
Space booking, noise monitoring, adaptive lighting

SCN-06 校园—城市开放门廊 Campus-City Porch G2

WHO 谁 师生、访客、社区居民  
Faculty, students, visitors, residents

WHERE 哪里 高校与城市界面门廊  
University Interface Porches

WHEN 何时 日间  
Daytime

SPACE 空间 AI+ AI+ 人流计数，开放时间建议，安防联动  
People counting, opening-hours advice, security linkage

SCN-07 低速配送服务湾 Low-Speed Delivery Bay G4

WHO 谁 快递员、骑手、居民  
Couriers, riders, residents

WHERE 哪里 大钟寺与AI原点边缘  
Edges of Dazhongsi & AI Origin

WHEN 何时 全天  
All day

SPACE 空间 AI+ AI+ 配送路径优化，停靠预约，充电调度  
Route optimization, slot booking, charging scheduling

SCN-09 站前会面客厅 Station Meeting Lounge G2 G3

WHO 谁 接送人群、旅客、居民  
Pick-up people, travelers, residents

WHERE 哪里 大钟寺站前广场与客厅  
Station Forecourt & Lounge

WHEN 何时 全天  
All day

SPACE 空间 AI+ AI+ 会面指引、拥挤度提示，公共事件信息  
Meeting guidance, crowd indicator, public event info

SCN-10 存量商业夜间客厅 Night Retail Living Room G3

WHO 谁 居民、游客、商户  
Residents, visitors, merchants

WHERE 哪里 AI原点与大钟寺既有商业轴  
Existing Retail Axes

WHEN 何时 夜间  
Night

SPACE 空间 AI+ AI+ 客流热力，商户运营建议，照明能耗优化  
Crowd heat map, merchant tips, lighting energy optimization

SCN-11 再生建筑创新展厅 Adaptive Reuse Innovation House G3 G4

WHO 谁 公众、创业者、学生  
Public, start-ups, students

WHERE 哪里 再生建筑与创新展厅  
Adaptive Reuse Buildings

WHEN 何时 日间 + 活动时段  
Daytime + Events

SPACE 空间 AI+ AI+ 展览导览、沉浸式互动、设备状态监测  
Exhibit guide, immersive interaction, equipment monitoring

SCN-12 无App京张文化步行 No-App Jingzhang Culture Walk G0 G1 G2

WHO 谁 游客、家庭、老年人  
Visitors, families, seniors

WHERE 哪里 京张文化步行轴  
Jingzhang Cultural Walk

WHEN 何时 全天  
All day

SPACE 空间 AI+ AI+ 离线语音导览，AR识别，无障碍提示  
Offline voice guide, AR recognition, accessibility prompts

场景覆盖维度 Scenario Coverage

通勤与换乘 Commuting & Transfer

学习与工作 Study & Work

休闲与社交 Leisure & Social

文化与旅游 Culture & Tourism

物流与服务 Logistics & Service

安全与韧性 Safety & Resilience

验证指标示例 Example Metrics

驻留时长 Dwell Time

步行量 Walking Volume

拥挤度 Crowding Index

满意度 Satisfaction

能耗 Energy Use

碳排放 Carbon Emission

说明 Notes

1. 场景可单独或组合实施，依赖于G0-G4厚基面与组件层的支撑。

2. AI为增强层，可替换、可迭代，不以算法替代公共性。

图例 Legend

人物 People

空间 Space

时间 Time

AI+ 增强 AI+ Enhancement

基线 Baseline

验证 Test

3 FLAGSHIP SCENARIOS

SCN-01 / TEST GARDEN

research + public garden  
Zhongzhiyuan / day

SCN-04 / SHARED STREET

learning + daily life  
AI Origin / dusk

SCN-08 / 4Q TRANSFER

access + station-city room  
Dazhongsi / night

9 SUPPORT SCENARIOS

workday courtyard  
blue-green walk  
night learning corner  
campus-city porch  
low-speed delivery bay  
station meeting lounge  
night retail living room  
adaptive reuse house  
no-app heritage walk

Every scenario requires a non-AI baseline.  
Equipment and data use remain to verify.

UNIFIED CARD LOGIC

WHO WHERE WHEN SPACE ACTION AI+ FALLBACK TEST

Validation focuses on accessibility, comfort, dwell, safety, satisfaction, energy and maintenance - without presuming a positive AI effect.

P14 | DESIGN SCENARIOS | AI IS OPTIONAL | PRIVACY + SAFETY + OPERATIONS + PERFORMANCE TO VERIFY



五类人物 × 三个公共地标 5 Personas × 3 Public Landmarks

不同的人，不同的时间，在厚基面上更好的城市生活  
Different People, Different Times, Better Urban Life on the Thick Ground

五类人物  
5 Personas

01 研究生 / 青年研究者  
Graduate Student / Young Researcher

WHO 谁: 18-28 岁, 研究、学习、参与创新项目  
18-28 yrs, research, study, innovation projects

NEED 需求: 安静学习、灵活交流、快速网络、便捷餐饮  
Quiet study, flexible exchange, fast network, convenient food

TIME 时间: 工作日白天、傍晚、考试/项目集中期  
Weekdays daytime & evening, exam / project periods

SPACE 空间 (偏好): 学习角落、共享庭院、口袋客厅  
Study corners, shared courtyards, pocket living rooms

THICK-GROUND RESPONSE  
• 连续慢行步行 + 静音角落  
• 连续网络 + 共享桌面  
• 高速Wi-Fi + 充电  
• 夜间安全照明

主要活动  
Key Activities

02 初创工程师 / 创业者  
Startup Engineer / Entrepreneur

WHO 谁: 22-35 岁, 创业团队、远程办公  
22-35 yrs, startup teams, remote work

NEED 需求: 团队讨论、原型测试、临时会议、资源对接  
Team discussion, prototyping, meeting, resource match

TIME 时间: 白天为主, 午休与傍晚交流活跃  
Mainly daytime, active at lunch & evening

SPACE 空间 (偏好): 开放庭院、共享庭院、测试花园、会议角  
Active ground floor, courtyard, test garden, meeting pods

THICK-GROUND RESPONSE  
• 共享工位 + 会议桌  
• 可移动家具 + 灵活插座  
• AI测试接口 + 数据可视化屏  
• 运动发单与资源共享平台

主要活动  
Key Activities

03 长期居住的老年居民  
Elderly Long-term Resident

WHO 谁: 60岁以上, 长期居住, 日常散步与社交  
60+ yrs, long-term resident, daily walk & social

NEED 需求: 无障碍通行、休息座椅、医疗与康养可达  
Barrier-free access, resting, healthcare & service

TIME 时间: 清晨与上午, 傍晚散步与锻炼  
Morning & late afternoon walks

SPACE 空间 (偏好): 连续步道、树荫停留带、社区客厅  
Continuous paths, shaded seating, community living

THICK-GROUND RESPONSE  
• 无障碍连续步行 + 缓坡  
• 树荫遮蔽 + 扶手  
• 清晰标识 + 一键呼叫  
• 冬季防风 + 避风空间

主要活动  
Key Activities

04 带儿童的社区家庭  
Family with Children

WHO 谁: 25-40 岁家庭, 孩子0-12岁  
25-40 yrs family, children 0-12 yrs

NEED 需求: 安全游乐、教育体验、亲子社交、便利服务  
Safe play, learning experience, parent-child social, convenience

TIME 时间: 周末与傍晚、假期全天  
Weekends & evenings, holidays all day

SPACE 空间 (偏好): 口袋公园、开放场地、学习院落  
Pocket parks, open fields, learning courtyards

THICK-GROUND RESPONSE  
• 安全慢行环境 + 低矮植被  
• 儿童游乐 + 自然教育  
• 母婴室 + 家庭卫生间  
• 近距服务与停车

主要活动  
Key Activities

05 首次到访的国内外访客  
First-time Visitor (Domestic / Intl.)

WHO 谁: 游客、商务出行者、短期停留者  
Visitors, business travelers, short-stay

NEED 需求: 清晰导视、重点与文化体验、便捷换乘  
Clear wayfinding, cultural experience, easy transfer

TIME 时间: 白天游览为主, 夜间体验部分场景  
Daytime visit mainly, some night experience

SPACE 空间 (偏好): 主步行廊、文化路径、换乘节点、服务点  
Main corridor, cultural path, transfer node, service point

THICK-GROUND RESPONSE  
• 多语言导视 + 数字地图  
• 文化故事 + 互动装置  
• 无App信息获取  
• 行李寄存 + 公共卫生间

主要活动  
Key Activities

厚基面层级 Thick Ground Levels (G0-G4)

G0 公共通行  
Movement

G1 生态停留  
Stay

G2 过渡空间  
Transition

G3 开放共享  
Active

G4 服务支撑  
Support

主要活动类型 Key Activity Types

通勤/换乘  
Commute

学习/工作  
Study / Work

休闲/社交  
Leisure / Social

服务/生活  
Service / Daily

文化/体验  
Culture / Experience

人物在三大重点区的使用偏好 (相对强度)  
Persona Usage Preference in Three Key Areas (Relative Intensity)

| 人物 Personas                               | 众智园<br>AI Test Garden | AI原点<br>Origin Shared Street | 大钟寺<br>Dazhongsì 4Q Node |
|-------------------------------------------|-----------------------|------------------------------|--------------------------|
| 01 研究生 / 青年研究者<br>Young Researcher        | ●●●                   | ●●●                          | ●                        |
| 02 初创工程师 / 创业者<br>Entrepreneur            | ●●                    | ●●●                          | ●●                       |
| 03 老年居民<br>Elderly Resident               | ●●                    | ●●                           | ●●●                      |
| 04 家庭 (带儿童)<br>Family with Children       | ●●●                   | ●●●                          | ●                        |
| 05 访客 (国内外)<br>Visitor (Domestic / Intl.) | ●                     | ●●                           | ●●●                      |

强 Strong ●●● 中 Medium ●● 弱 Weak ● 很弱 Very Weak ○

三个公共地标 3 Public Landmarks

LM-01 百年时标 Century Gauge

京张遗产的时间标记与公共阅读空间  
A time marker and public reading space of the Jingzhang heritage

设计要点 Key Features

- 线性节点 Linear nodes
- 可读可读 Site & Read
- 历史故事 Heritage Story
- 夜间照明 Night Lighting

位置示意 Location

LM-02 原点开放门廊 Origin Open Porch

开放的学习与交换界面  
An open porch for learning and exchange

设计要点 Key Features

- 开放庭院 Open Ground Floor
- 门廊空间 Porch & Threshold
- 共享学习 Shared Learning
- 灵活可变 Flexible Use

位置示意 Location

LM-03 大钟寺城市交换厅 Dazhongsì Urban Exchange Hall

站城连接的会面、换乘与服务中心  
A station-city hub for meeting, transfer and public service

设计要点 Key Features

- 换乘集散 Transfer Hub
- 雨棚与庇护 Shelter
- 信息交流 Information Exchange
- 便民服务 Public Services

位置示意 Location

厚基面支撑要素 Thick Ground Support Elements

连续步行  
Continuous Walk

无障碍  
Accessible

绿荫与生态  
Shade & Ecology

座椅与停留  
Seating & Stay

安全与照明  
Safety & Lighting

导航与信息  
Wayfinding & Info

数字赋能  
Digital Enablement

低碳与韧性  
Low-carbon & Resilient

验证指标示例 Example Metrics

步行连通率  
Walkability %

座椅可达率  
Seat Accessibility

无障碍覆盖率  
Accessibility Coverage

日均活动人数  
Daily Active Users

公众满意度  
Public Satisfaction

说明 Notes

- 人物需求与空间响应基于场景研究与实地观察总结。
- 地标为概念设计, 位置与规模可根据后续规划与实施条件优化。

1. Needs and responses are summarized from scenario research and field observation.  
2. Landmark designs are conceptual and adaptable to planning and implementation.

图例 Legend

主要人群  
Persona

主要空间  
Space

时间  
Time

AI+增强  
AI+ Enhancement

PERSONA-TO-SPACE TEST

continuous walk      shade + rest      clear wayfinding      accessible service      safe evening use

Each prototype is reviewed through different bodies, times, abilities and reasons for being in the city.

FIVE PUBLICS

01 / young researcher  
02 / entrepreneur  
03 / older resident  
04 / family with children  
05 / first-time visitor

THREE LANDMARKS

LM-01 / CENTURY GAUGE

walk + sit + read history  
heritage as usable public space

LM-02 / OPEN PORCH

learn + meet + exchange  
threshold as shared institution

LM-03 / EXCHANGE HALL

arrive + transfer + receive help  
station forecourt as urban room

INCLUSION RULE

Basic navigation, comfort and public service must work without an app.

P15 | DESIGN PERSONAS + LANDMARKS | INCLUSIVE BASELINE | CO-DESIGN + ACCESSIBILITY REVIEW TO VERIFY



1 / FOUR EVIDENCE STATES

- SOURCE

verified external evidence or official data
- DESIGN

project proposition or spatial action
- PROVISIONAL

concept geometry or unresolved boundary
- TO VERIFY

requires official GIS or technical input

**RULE** Unverified numbers are removed, not cosmetically qualified.  
The asset registry records risk, retained content, verification needs and next action.

2 / METRIC PROTOCOL / NO CLAIMED VALUES

- ACCESS

continuity + barriers
- COMFORT

shade + shelter + dwell
- INCLUSION

abilities + no-app use
- PUBLIC LIFE

dwell + mix + satisfaction
- OPERATIONS

maintenance + staffing
- REVERSIBILITY

fallback + removal

**BEFORE / DURING / AFTER**  
Baseline observation - pilot monitoring - public review - adaptation.  
Targets and methods require a later measurement plan and accountable data governance.

3 / CONDITIONAL PHASING

- L1

LIGHT TOUCH

repair, seating, shade, signs, pilots
- L2

INTERFACE RETROFIT

porch, ground floor, courtyard, crossing
- L3

STRUCTURAL CHANGE

reuse, station-city work, selective new
- ADVANCE ONLY IF

public value confirmed

responsibility assigned

technical risk resolved

operations funded

feedback supports scale

No fixed dates. Sequence follows evidence, consent, approvals and operating readiness.

4 / SUBMISSION + RESPONSIBILITY BOUNDARY

- READY AS DESIGN PROPOSAL

method, prototypes, components, scenarios
- PROVISIONAL

boundaries, alignments, node locations
- NOT TO SCALE

sections, details and equipment interfaces
- NOT CLAIMED

area, capacity, cost, traffic, performance
- NOT IMPLIED

approval, demolition or operational authority

FINAL PRINCIPLE

A strong proposal can be ambitious in design and precise about the limits of its evidence.  
AI-assisted imagery communicates design intent; it does not replace official GIS, survey, engineering, consent or governance.